

# analogue-risk: a fast, leakage-safe engine for regime-conditioned analogue forecasting of financial risk

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## Abstract

Nearest-neighbour ("analogue") forecasting is attractive in finance because every forecast can be traced to concrete historical episodes, yet published evaluations are often vulnerable to look-ahead bias, overlapping samples and backtest overfitting. **analogue-risk** is an open-source Rust engine with Python bindings that makes such evaluations fast and leakage-safe. It provides regime-conditioned  $k$ -nearest-neighbour and Gaussian-kernel analogue forecasters over arbitrary causal features and targets, an exclusion zone that keeps analogue outcome windows independent, a Markov-mixture over volatility regimes, a trade simulator shared between label construction and backtesting, Hawkes and GARCH likelihoods, and a complete walk-forward evaluation suite: Diebold–Mariano tests, the model confidence set, VaR/ES backtests, deflated Sharpe ratios and pre-registered hypothesis tests. A full-history analogue search over 200,000 bars runs in about 90 seconds on a laptop, and every table of the accompanying study is reproduced by a single command.

*Keywords:* analogue forecasting, nearest neighbours, volatility forecasting, value-at-risk, Hawkes process, backtesting, Rust, reproducibility

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## Code metadata

| Nr. | Code metadata description                                         |                                                                                                                                                   |
|-----|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| C1  | Current code version                                              | v2.0.0                                                                                                                                            |
| C2  | Permanent link to code/repository                                 | <a href="https://github.com/ManasDasri/analogue-risk">https://github.com/ManasDasri/analogue-risk</a>                                             |
| C3  | Permanent link to reproducible capsule                            | –                                                                                                                                                 |
| C4  | Legal code license                                                | MIT                                                                                                                                               |
| C5  | Code versioning system used                                       | git                                                                                                                                               |
| C6  | Software code languages, tools and services used                  | Rust, Python; PyO3, maturin, rayon; NumPy, SciPy, pandas, scikit-learn, Matplotlib                                                                |
| C7  | Compilation requirements, operating environments and dependencies | Python $\geq 3.12$ (prebuilt wheels: Linux, macOS, Windows); building from source needs Rust stable                                               |
| C8  | Link to developer documentation/manual                            | <a href="https://github.com/ManasDasri/analogue-risk/blob/main/docs/api.md">https://github.com/ManasDasri/analogue-risk/blob/main/docs/api.md</a> |
| C9  | Support email for questions                                       | bl.sc.u4cse24229@bl.students.amrita.edu                                                                                                           |

## 1. Motivation and significance

The method of analogues [1, 2] forecasts the future of a system by finding past states that resemble the present and aggregating what happened next. In finance it has been used for exchange rates [3] and is widely re-implemented in retail trading tools, where it appeals because it is interpretable: every forecast is backed by a list of historical episodes a user can inspect. The same features make analogue methods easy to evaluate wrongly. Three failure modes are common: (i) *look-ahead*: an analogue whose outcome window has not yet closed at the forecast time leaks the future; (ii) *dependent neighbours*: adjacent bars have nearly identical features, so the “ $k$  nearest” analogues are often one episode counted  $k$  times, which makes confidence statements meaningless; and (iii) *backtest overfitting*: many configurations are tried and the best is reported [4, 5].

**analogue-risk** grew out of a TradingView strategy that combined a one-dimensional analogue matcher, a Markov volatility regime, a Poisson crash gate and Kelly sizing. Rebuilding it to be evaluated properly required an engine that is causal by construction, fast enough to search the *entire* history at every bar, and paired with the statistical tooling that forecasting and

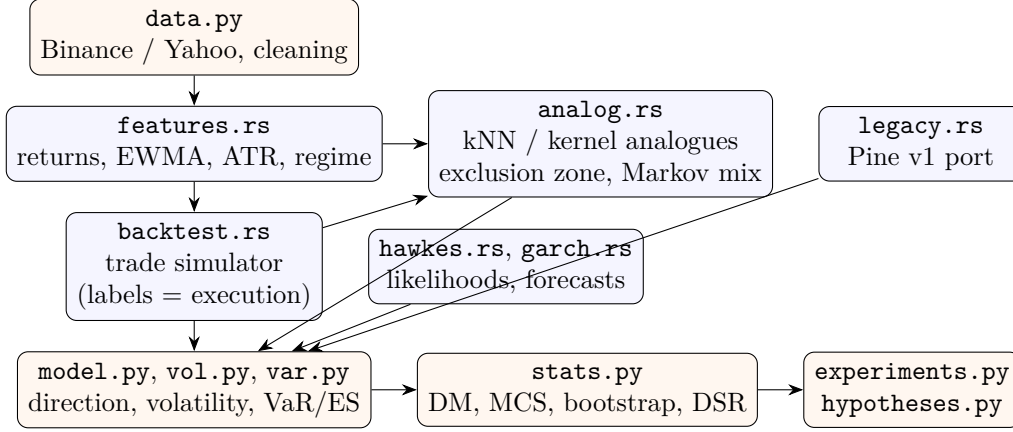


Figure 1: Architecture. Blue: Rust core; orange: Python layer.

finance journals expect. No existing open-source package combines these. The software makes three contributions: a leakage-safe, parallel analogue engine; a label/execution design in which the same trade simulator scores historical analogues and live trades; and an end-to-end, pre-registrable evaluation pipeline. The accompanying study [6] uses it to show, across 24 assets, that analogue methods carry no directional information but produce well-calibrated, competitive risk forecasts.

## 2. Software description

### 2.1. Software architecture

The package is a Rust crate compiled to a Python extension module (`analogue_risk._core`) with PyO3 and maturin, plus a thin Python layer (Fig. 1). All per-bar work is in Rust and releases Python’s global interpreter lock; per-bar forecasts are independent and run in parallel with rayon.

### 2.2. Software functionalities

*Analogue engine.* At bar  $t$  a causal embedding  $x_t \in \mathbb{R}^m$  is compared with every past anchor  $j$  whose targets  $y_j \in \mathbb{R}^p$  are fully observed ( $j \leq t - \delta$  for a declared delay  $\delta$ ). In  $k$ NN mode the nearest anchors are accepted greedily, per volatility regime, subject to an exclusion zone ( $|j - j'| \geq e$  for all accepted  $j, j'$ ), so outcome windows of length  $h \leq e$  never overlap. A partial selection (`select_nth_unstable`) followed by a full sort only when needed keeps this near  $O(n)$  per bar. Regime groups are combined with the  $h$ -step forecast

of a two-state Markov chain estimated from prefix transition counts in  $O(1)$  per bar, giving  $\hat{y}_t = \sum_s \pi_s(t) \bar{y}_s(t)$  with standard errors of the mixture mean. A Gaussian-kernel mode (Nadaraya–Watson [7, 8]) weights all candidates and reports Kish effective sample sizes [9]. `neighbours()` returns the index and weight of every analogue, so any functional of the analogue distribution (quantiles, expected shortfall, explanations such as Fig. 2) can be computed. A configuration guard rejects settings in which the exclusion zone would force non-neighbours into the set.

*Labels equal execution.* `backtest.rs` simulates a trade (next-open fill, intrabar stop-before-target, gap fills, time stop, costs). The same function computes the R-multiple label of every historical anchor, so a forecaster is trained on exactly the event the strategy executes; a test asserts bit-level agreement.

*Benchmarks and risk models.* EWMA [10], GARCH(1,1) [11] by maximum likelihood, HAR [12], gradient boosting [13, 14], exponential Hawkes processes [15] with an exact  $O(n)$  likelihood, historical and filtered historical simulation [16], and analogue-conditioned VaR/ES.

*Evaluation.* Walk-forward folds in which every fitted quantity sees only earlier data; QLIKE [17]; Diebold–Mariano tests [18] with Newey–West variance [19]; the model confidence set [20] with the stationary bootstrap [21]; Kupiec [22] and Christoffersen [23] coverage tests and the Fissler–Ziegel FZ0 loss [24, 25]; probabilistic and deflated Sharpe ratios [26, 4]; and a pre-registered hypothesis module with Holm correction [27].

*Public API and data.* `Analogue` exposes the engine with named, validated options and labelled outputs; `volatility_forecast(prices, horizon)` is a one-call risk forecaster. Every series used in the study is pinned (last bar, row count, SHA-256), so re-runs reproduce the published numbers or warn that a source has changed; Binance data fall back to the public bulk archive where the API is geo-blocked.

*Quality assurance.* Thirty-six Python and twelve Rust tests cover causality (forecasts, labels and the legacy port are identical when all data after  $t$  is removed), likelihoods against brute force, Hawkes parameter recovery, recovery of a planted signal versus a random walk, known-answer checks of every backtest and statistic, and a regression test that reproduces the pre-registered hypothesis tables. CI runs them on Linux, macOS and Windows

with Python 3.12 and 3.13 and enforces formatting; tagged releases build wheels for all three platforms. An executable harness compares the legacy port with TradingView’s own trade list.

### 3. Illustrative examples

Listing 1 forecasts next-day volatility of the S&P 500 with analogues and compares it with HAR and GARCH under the walk-forward protocol.

Listing 1: Walk-forward volatility forecasting.

```
from analogue_risk import data, model as M, vol as V, stats as S
from analogue_risk.experiments import split
mkt = M.Market(data.yahoo("^GSPC", start="1927-12-30"))
d = V.VolData(mkt, V.DAILY_W) # h = 5 days
cal, folds = split(mkt.n, V.DAILY_W.warmup)
F, _ = V.walk_forward(d, folds) # EWMA, GARCH, HAR, GBM, Analogue,
    Analogue+HAR
ok = (d.realised > 0)
ok[:cal] = False
losses = {k: V.qlike(d.realised[ok], f[ok]) for k, f in F.items()}
print(S.model_confidence_set(losses, B=1000, block=10))
```

Three executed tutorials accompany the code: volatility forecasting with explanations, VaR backtesting, and—to show the engine is not specific to finance—forecasting the Lorenz system with delay-embedding analogues, where it beats persistence by 72–94% in RMSE depending on the horizon. Figure 2 shows the interpretability the engine exposes: for a given hour it returns the historical volatility paths closest to the present and what followed them.

*Performance.* On an Apple M4 (10 cores) the capped-history  $k$ NN forecaster processes about 30,000 bars/s; the full-history search is  $O(n^2)$  and forecasts 200,000 bars in about 90 s; a backtest over 76,000 bars takes about 3 ms. The complete development study (9 assets, 15 series–horizon pairs, bootstrap inference) runs in about 35 minutes.

### 4. Impact

The engine turned a retail-style trading indicator into a falsifiable research instrument. Its first use [6] (i) replicated the original strategy bar-for-bar and

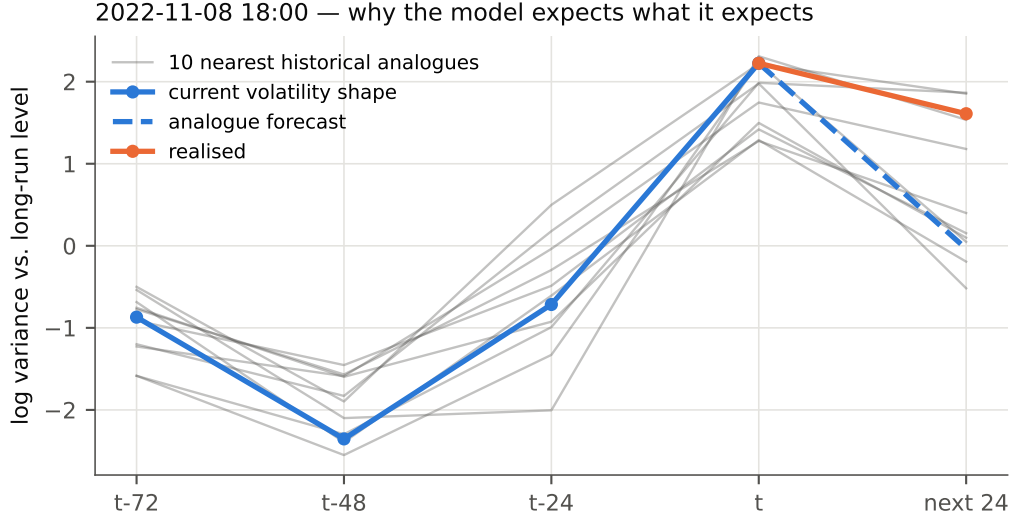


Figure 2: Explaining a forecast: BTC on 8 November 2022. The current volatility shape (blue), its ten nearest historical analogues (grey) and what followed them, the forecast, and the realised outcome (orange).

showed that its reported edge does not survive out-of-sample testing with realistic costs; (ii) showed that analogue forecasts of volatility, combined with HAR, are statistically indistinguishable from the best standard models, that kernel weighting improves them by 5%, and that analogue-conditioned VaR is as well calibrated as filtered historical simulation; and (iii) did so under a public pre-registration, with the confirmatory analysis code committed before the data it analyses. All eight pre-registered hypotheses replicated on fifteen assets never used in development. Because targets and embeddings are arbitrary arrays, the engine applies unchanged to other analogue problems (other assets, realised measures, macroeconomic series), and its leakage-safe design and statistical suite lower the cost of evaluating such methods honestly. Researchers can reproduce every number with one command, and students can inspect every forecast’s analogues.

## 5. Conclusions

**analogue-risk** provides a fast, causal and statistically complete toolkit for analogue forecasting in finance. Planned extensions are multivariate

(cross-asset) analogues, realised measures from intraday data, and learned combination weights.

### **Declaration of competing interest**

The authors hold a patent on the original trading method from which this software was derived. The software is released under the MIT license.

### **Declaration of generative AI and AI-assisted technologies in the writing process**

During the preparation of this work the authors used Claude (Anthropic) to assist with software development and drafting of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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